

3.5.4**a.**Sum of digits is 9 means that $9 \mid n$

$$\frac{9999-9}{9} + 1 = \boxed{1111}$$

b.Four numbers add up to 9, $\boxed{\frac{(9+3)!}{3!9!}}$ ways to have numbers add up to 9The numbers can be zero, since $0001 = 1$ is valid!**c.**Four numbers that add up to 18, so there must be $\boxed{\binom{18+3}{3}}$ ways**3.6.5**

Multiple of 99, so multiple of 11 and 9 at the same time

Starts at 1089, goes up to 9999

 $abcd$ is divisible by 99 if $11 \mid (a - b + c - d)$ and $9 \mid (a + b + c + d)$ $x = a + c$, and $y = b + d$. Then $11 \mid (x - y)$, and $9 \mid (x + y)$ $x - y = \{0, 11\}$ - maximum since there's 4 digits only

$$x + y = \{9, 18, 27\}$$

$$2x = \{9, 18, 27, 20\}$$

$$2y = \{9, 18, 27, 7, 16\}$$

Actually, the possible values for $2x$ and $2y$ cannot be odd!

$$2x = \{18, 20\}$$

$$2y = \{18, 16\}$$

$$x = \{9, 10\}$$

$$y = \{9, 8\}$$

Checking against the initial conditions for $x - y$ and $x + y$, $x = 9$ and $y = 9$ otherwise it won't work

How do I solve this without bashing...